Photometric Test Report

Reference Standards

UL1598-2008
ANSI C82.77-10-2014
IES LM-79-2019

Prepared For

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Test Laboratory:
UL-CCIC Company Limited
Test Laboratory Address:
No.2, Chengwan Road, Suzhou Industrial Park, Suzhou 215122, China
Catalog Number
BPANEL-75S5-UNV-D-8SC61-24-G-WH-P

Project Number
4791498147.3
Report Number
4791498147.3-3a

Test Date
2024-10-30~2024-11-07
Issue Date
2024-11-21
Revision Date
N/A

Prepared By	Approved By
Becky Fan	Roger Xu

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Test List

Sample Received Date: 2024-10-16

Test Item	Sample ID	Model Number	Tests Conducted By
Integrating Sphere Test	7712178	BPANEL-75S5-UNV-D-8SC61-24-G-WH-P	Hom Dong
Goniophotometer Test	7712178	BPANEL-75S5-UNV-D-8SC61-24-G-WH-P	Hom Dong
THD and PF Test	7712178	BPANEL-75S5-UNV-D-8SC61-24-G-WH-P	Hom Dong
In-Situ Temperature Measurement Test	7712178	BPANEL-75S5-UNV-D-8SC61-24-G-WH-P	Hom Dong

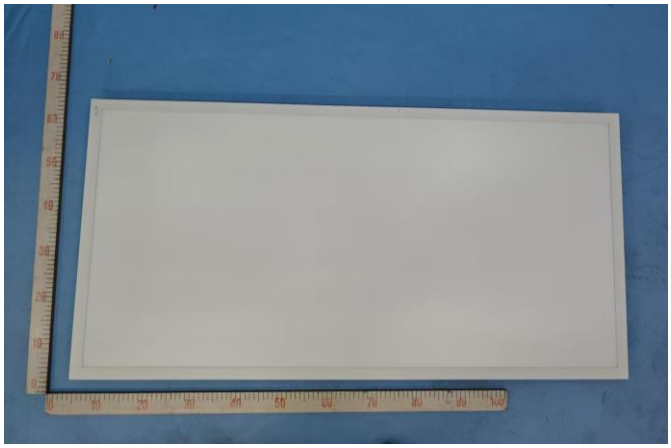
Remark (if any)

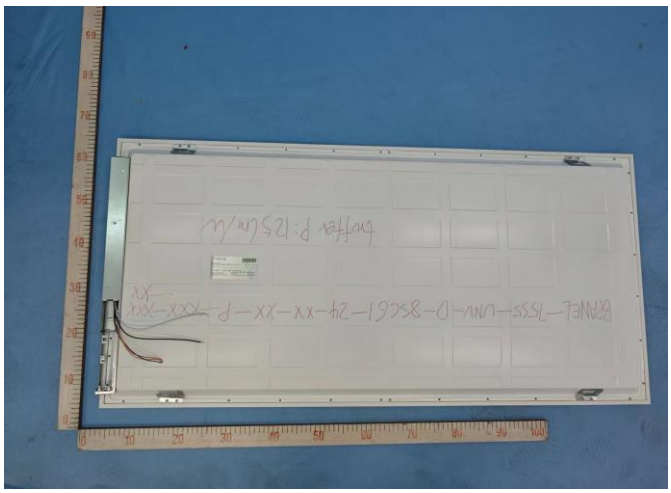
1. UL test equipment information is recorded on Meter Use in UL's Aurora database.

Product Description

Luminaire Description: 2x4 Luminaires for Ambient Lighting of Interior Commercial Spaces
Model Number: BPANEL-75S5-UNV-D-8SC61-24-G-WH-P
Electrical Ratings and CCT: 120-277Vac, 50/60 Hz, 75W max, 3000K/3500K/4000K/5000K/5700K/6500K tunable
Driver Model Number: BQ-65L6-1470-44-M56
LED Package: BXVN-30E-13H-9NV, BXVN-65E-13H-9NV, Bridgelux Inc
Family Model and Variation: BPANEL-75S5-UNV-D-8SC61-24-XX-YY-P-ZZZ-***-##, where "XX" represent mounting, can be G=Grid mounting, SF=Surface mounting, SP=Suspended mounting, DF=Drywall flange mounting. "YY" represent Finish color, can be WH=White or others. "ZZZ" represent Emergency, can be EM*= Emergency types(* can be blank or any letter), blank=No emergency. "***" represent Sensor receptacle, can be SR#= Sensor receptacle types(# can be blank or any number), blank= Without sensor receptacle. ## represent junction box, can be JB=with junction box, blank=without junction box.

Photos of Products Characteristics





Integrating Sphere Test

Model No.	BPANEL-75S5-UNV-D-8SC61-24-G-WH-P	Sample ID.	7712178
Operate time (Min.)	55	Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric paramters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 °C ± 0.9 °C.The reference standard lamp is power 100W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
3.The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

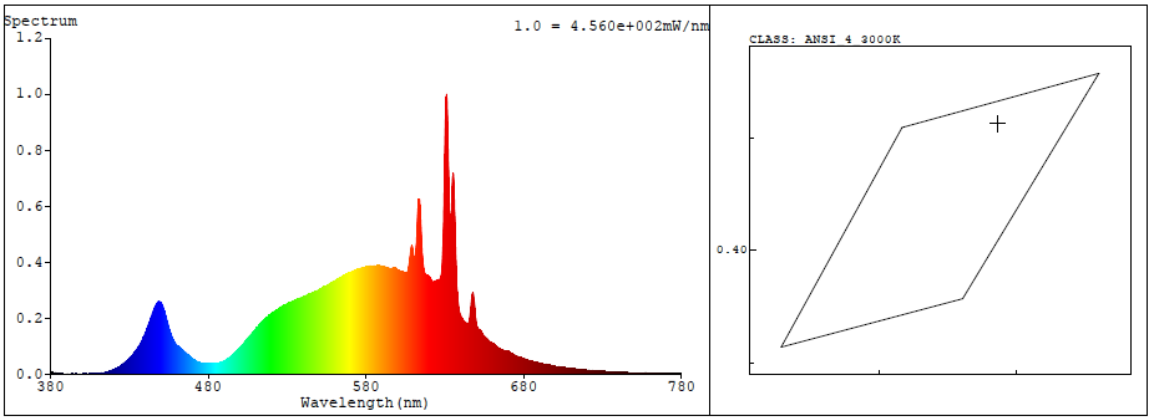
Integrating Sphere Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
25.1	120.1	60	0.593	70.79	0.9933	Horizontal

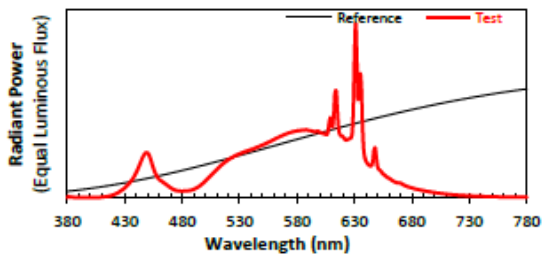
Test Results

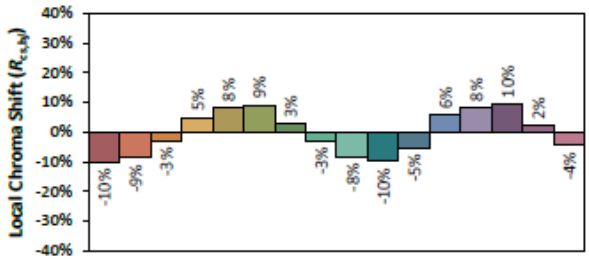
CCT (K)	CRI (Ra)	R9	Rf	Rg	Flux (lm)	Luminous Efficacy (lm/W)
3029	80	17	81	99	9422	133.1

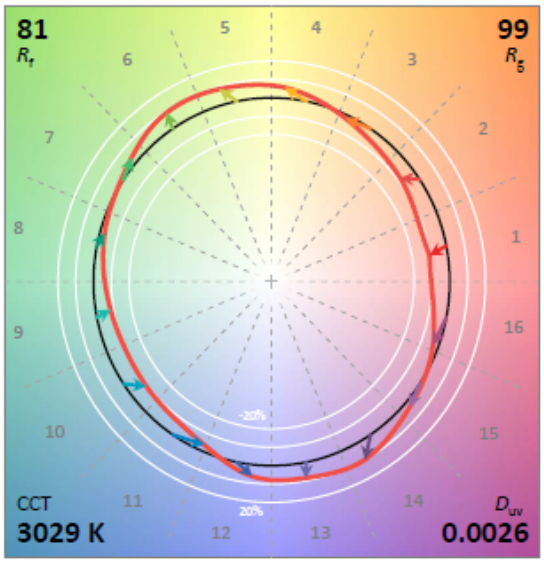
Spectrum

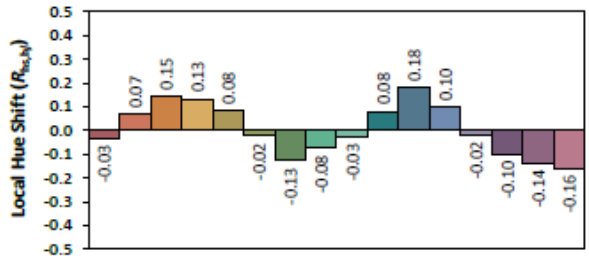


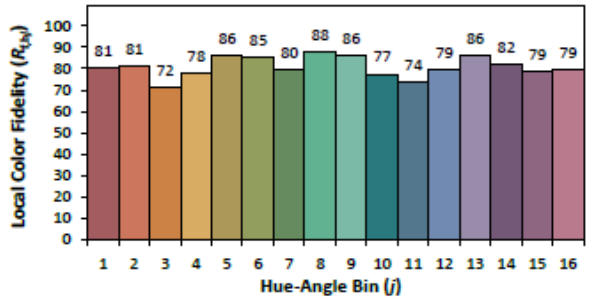
ANSI/IES TM-30-18 Color Rendition Report

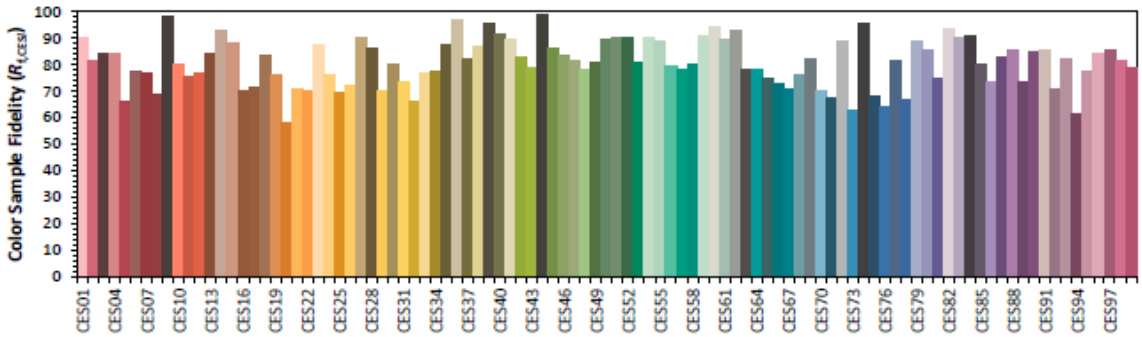












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.4386
y	0.4111
u'	0.2486
v'	0.5244

CIE 13.3-1995 (CRI)	
R_a	80
R_9	17

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

Integrating Sphere Test

Model No.	BPANEL-75S5-UNV-D-8SC61-24-G-WH-P		Sample ID.	7712178
Operate time (Min.)	55	Stabilization time (Min.)		50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric paramters were measured using an integrating sphere, a spectroradiometer and software. The ambient temperature condition inside the sphere was maintained at 25 °C ± 0.9 °C.The reference standard lamp is power 100W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
3.The sample measurements were made using a spectroradiometer connected by a fiber optic cable and detector through the detector port of the integrating sphere. Coating reflectance of the integrating sphere was 90% to 98%. Photometric measurement conditions was using 4π geometry. The self-absorption factor is applied in the final test result. The sample was operated at rated voltage and was stabilized before measurement. Chromaticity coordinates, correlated color temperature and color rendering index were calculated from the spectral radiant flux measurements taken at 1 nm intervals over the range of 380 to 780 nm.

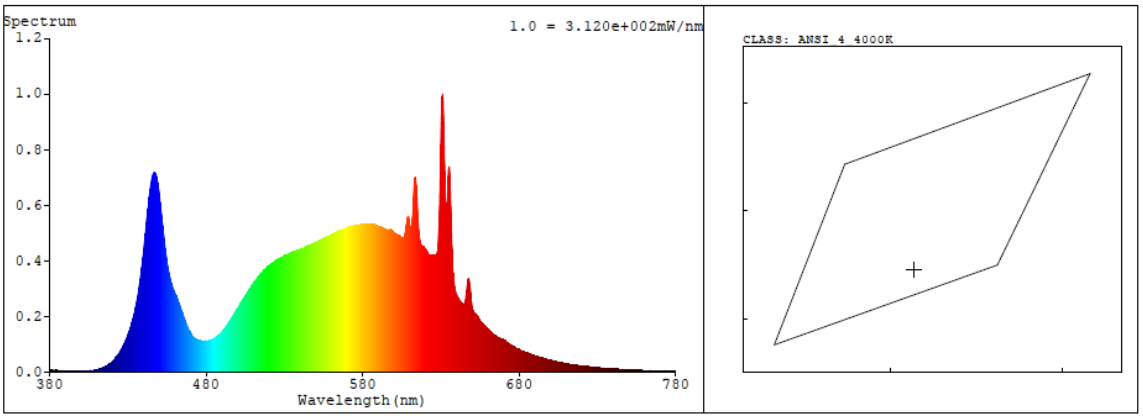
Integrating Sphere Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
25.1	120	60	0.590	70.34	0.9931	Horizontal

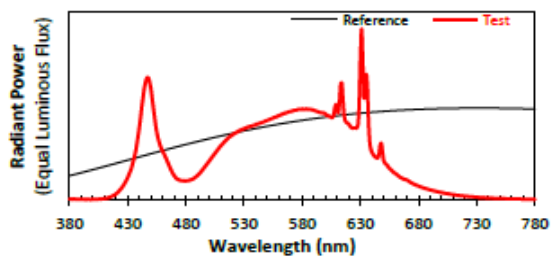
Test Results

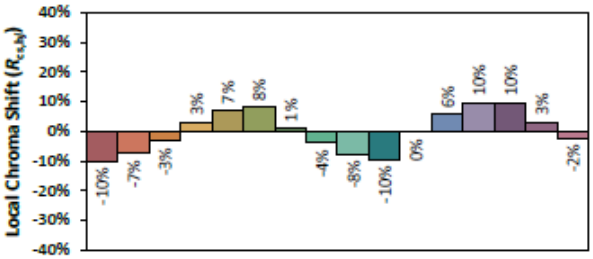
CCT (K)	CRI (Ra)	R9	Rf	Rg	Flux (lm)	Luminous Efficacy (lm/W)
3960	83	23	82	100	9458.5	134.5

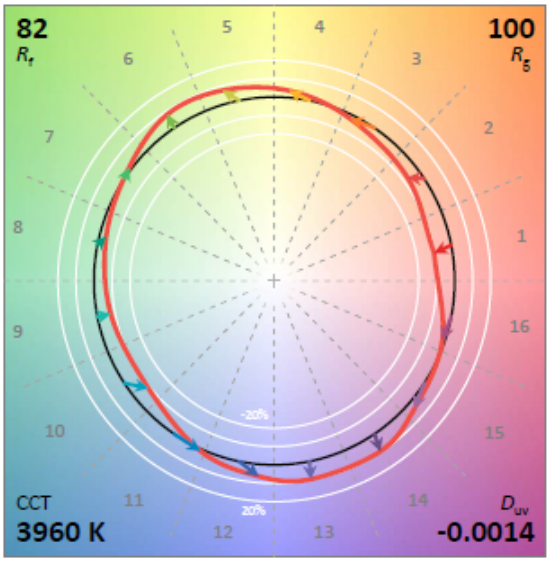
Spectrum

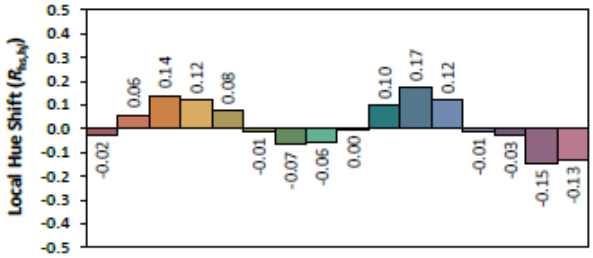


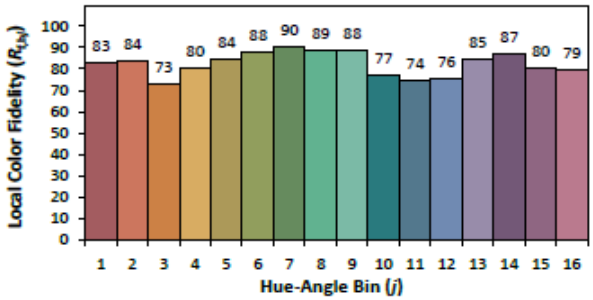
ANSI/IES TM-30-18 Color Rendition Report

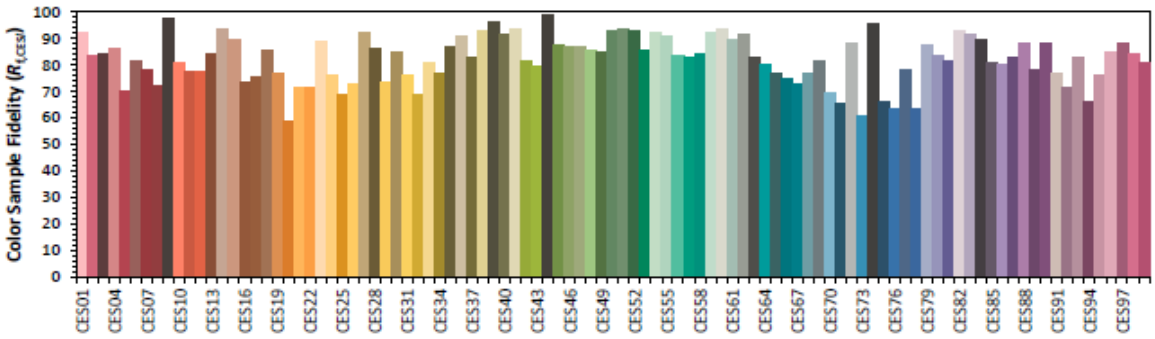












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3813
y	0.3744
u'	0.2266
v'	0.5007

CIE 13.3-1995 (CRI)	
R_a	83
R_9	23

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Integrating Sphere Test

Model No.	BPANEL-75S5-UNV-D-8SC61-24-G-WH-P	Sample ID.	7712178
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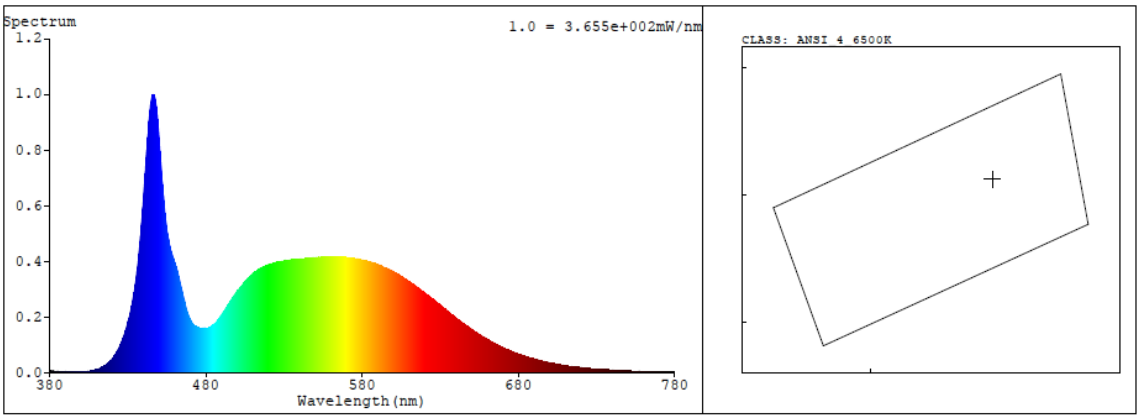
Integrating Sphere Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
25.1	120.1	60	0.592	70.63	0.9933	Horizontal

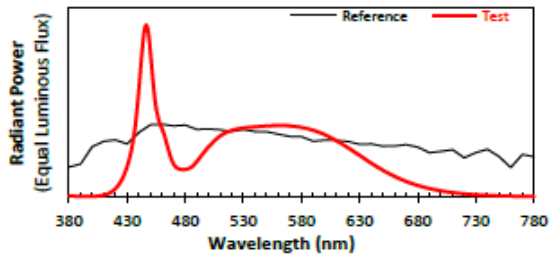
Test Results

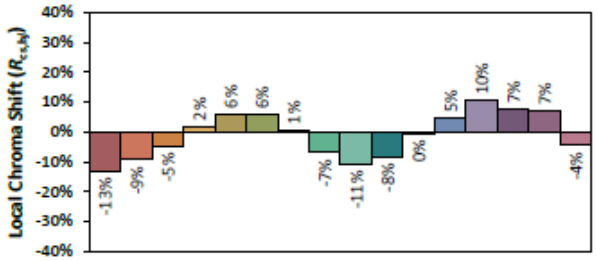
CCT (K)	CRI (Ra)	R9	Rf	Rg	Flux (lm)	Luminous Efficacy (lm/W)
6378	80	0	81	97	9438.2	133.6

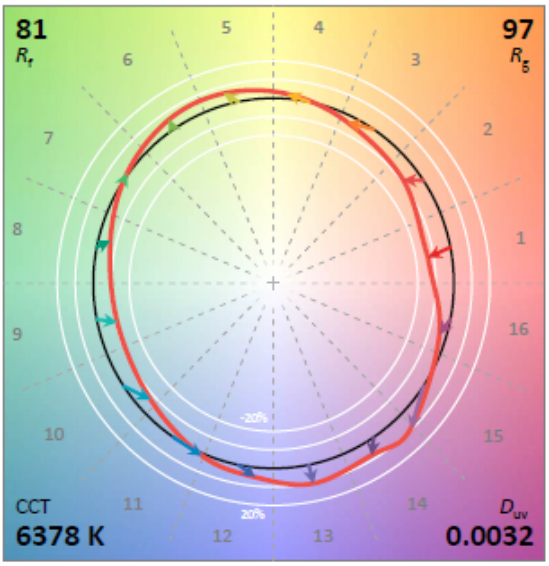
Spectrum

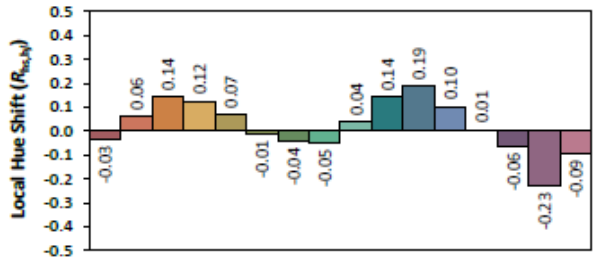


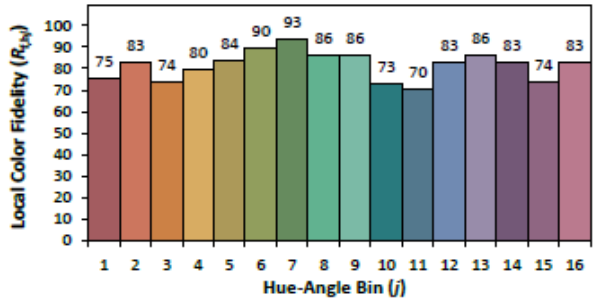
ANSI/IES TM-30-18 Color Rendition Report

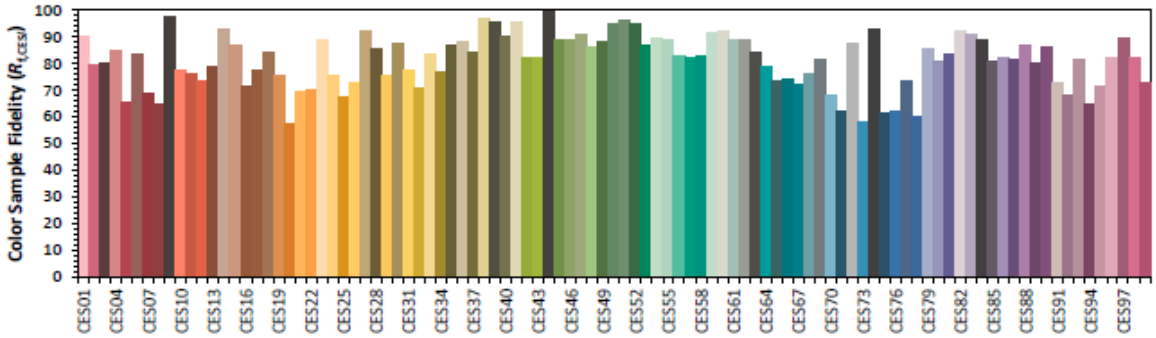












Notes: This is a recommended method for displaying ANSI/IES TM-30-18 information.

x	0.3148
y	0.3310
u'	0.1985
v'	0.4697

CIE 13.3-1995 (CRI)	
R_a	80
R_9	0

Colors are for visual orientation purposes only. Created with the ANSI/IES TM-30-18 Calculator Version 2.00.

Goniophotometer Test

Model No.	BPANEL-75S5-UNV-D-8SC61-24-G-WH-P	Sample ID.	7712178
Operate time (Min.)	65	Stabilization time (Min.)	50

Test Method

1.The sample was tested according to the IES LM-79-2019, and the product is assume to be brand new without seasoning.
2.Photometric parameters were measured using a type C goniophotometer and software.
3.The ambient temperature shall be maintained at 25 °C ± 0.9 °C, measured at a point not more than 1.5 m from the sample and at the same height as the sample. The reference standard lamp is power 400W omni-directional Incandescent lamp and was calibrated by National Institute of Metrology, China.
4.The samples were operated at rated voltage and was stabilized before measurement. Luminous flux, luminaire efficacy, zonallumen were calculated from the software taken at 1° vertical intervals and 22.5° horizontal intervals. Photometric distance was more than five times of the largest dimension of the test SSL product.

Goniophotometer Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
24.3	120	60	0.581	69.46	0.9955	Horizontal

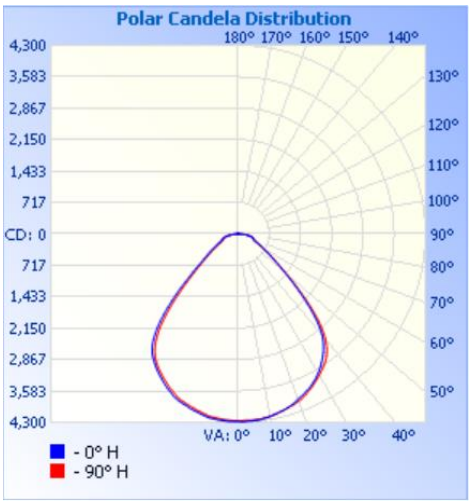
Test Results

Zonal Lumen Requirement (0°-60°)	Beam Angle (50%)		Spacing Criteria (0-180°)	Spacing Criteria (90°-270°)	Flux (lm)	Luminous Efficacy (lm/W)
	Horizontal Spread	Vertical Spread				
86.6%	90	90.7	1.28	1.30	9260.7	133.3

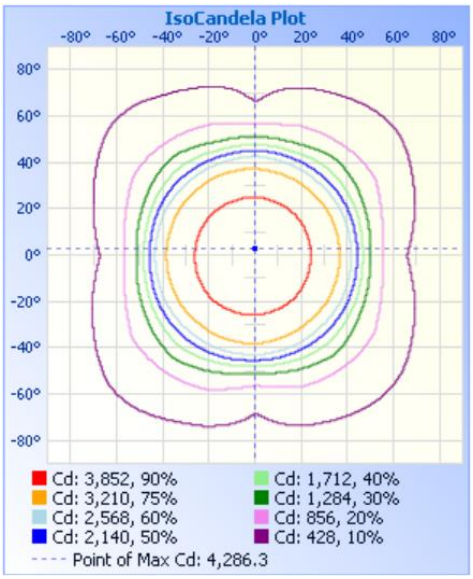
UGR
18.8

Goniophotometer Test (Cont'd)

Polar Candela Distribution



IsoCandela Plot



Goniophotometer Test (Cont'd)

Zonal Lumen Summary

Zonal Lumen Summary		
Zone	Lumens	% Luminaire
0-30	3,350.4	36.2%
0-40	5,443.7	58.8%
0-60	8,019.9	86.6%
60-90	1,219.5	13.2%
70-100	600.4	6.5%
90-120	8.9	0.1%
0-90	9,239.4	99.8%
90-180	21.2	0.2%
0-180	9,260.7	100%

Lumens Per Zone

Lumens Per Zone					
Zone	Lumens	% Total	Zone	Lumens	% Total
0-10	405.3	4.4%	90-100	5.2	0.1%
10-20	1,166.5	12.6%	100-110	1.8	0%
20-30	1,778.6	19.2%	110-120	1.9	0%
30-40	2,093.2	22.6%	120-130	2.2	0%
40-50	1,633.0	17.6%	130-140	2.6	0%
50-60	943.3	10.2%	140-150	2.7	0%
60-70	624.4	6.7%	150-160	2.3	0%
70-80	418.7	4.5%	160-170	1.8	0%
80-90	176.5	1.9%	170-180	0.8	0%

Goniophotometer Test (Cont'd)

Intensity Data(cd)

	0	22.5	45	67.5	90	112.5	135	157.5	180	202.5	225	247.5	270	292.5	315	337.5	360
0	4272	4272	4272	4272	4272	4272	4272	4272	4272	4272	4272	4272	4272	4272	4272	4272	4272
1	4286	4274	4265	4255	4259	4261	4270	4278	4279	4271	4264	4255	4258	4258	4273	4282	4280
2	4283	4272	4263	4258	4253	4264	4274	4274	4279	4271	4264	4257	4255	4264	4269	4278	4285
3	4276	4272	4266	4260	4259	4263	4276	4280	4279	4274	4263	4257	4251	4260	4269	4277	4276
4	4279	4271	4264	4260	4260	4261	4274	4277	4274	4269	4253	4247	4249	4256	4261	4269	4271
5	4272	4264	4260	4254	4254	4265	4268	4271	4270	4257	4249	4242	4237	4243	4254	4261	4268
6	4268	4256	4258	4246	4251	4259	4263	4266	4263	4254	4240	4232	4233	4235	4246	4252	4262
7	4265	4252	4250	4245	4249	4254	4255	4260	4258	4241	4235	4229	4224	4228	4242	4251	4255
8	4245	4246	4242	4244	4242	4245	4257	4252	4250	4240	4228	4215	4218	4225	4232	4240	4244
9	4237	4239	4240	4228	4237	4244	4249	4249	4248	4231	4218	4208	4211	4211	4225	4235	4240
10	4221	4227	4225	4222	4225	4230	4239	4237	4229	4219	4207	4193	4191	4196	4212	4221	4230
11	4209	4210	4209	4206	4211	4215	4220	4219	4214	4203	4187	4172	4171	4177	4191	4199	4208
12	4189	4192	4191	4191	4197	4201	4203	4202	4190	4178	4166	4157	4148	4156	4171	4184	4188
13	4173	4172	4173	4176	4175	4186	4193	4181	4174	4161	4143	4131	4134	4138	4151	4157	4170
14	4162	4154	4162	4160	4164	4166	4169	4170	4154	4137	4126	4116	4108	4121	4129	4140	4149
15	4145	4132	4136	4137	4144	4154	4153	4144	4140	4118	4104	4093	4091	4095	4108	4120	4126
16	4125	4114	4122	4126	4129	4136	4138	4129	4119	4102	4085	4073	4072	4078	4086	4097	4112
17	4100	4094	4100	4111	4108	4120	4117	4105	4093	4081	4063	4052	4054	4056	4065	4075	4091
18	4075	4079	4082	4091	4100	4099	4099	4092	4077	4058	4047	4031	4030	4039	4049	4057	4063
19	4051	4057	4070	4074	4078	4087	4080	4070	4059	4038	4025	4013	4016	4017	4026	4040	4053
20	4010	4039	4047	4061	4060	4065	4063	4048	4036	4023	4002	3989	3990	3995	4004	4013	4026
25	3872	3873	3885	3899	3907	3901	3900	3886	3867	3850	3827	3809	3812	3815	3825	3836	3851
30	3670	3665	3687	3710	3724	3720	3701	3685	3662	3629	3599	3595	3605	3601	3603	3624	3647
35	3389	3428	3435	3489	3507	3486	3442	3436	3419	3349	3284	3305	3341	3324	3301	3353	3404
40	2994	2937	2915	3014	3109	3006	2900	2946	2971	2802	2660	2716	2804	2736	2681	2816	2967
45	2242	2148	2152	2261	2357	2271	2187	2205	2156	2041	1955	1930	1945	1966	1965	2042	2139
50	1468	1433	1586	1550	1511	1577	1648	1521	1359	1428	1477	1311	1229	1357	1465	1374	1363
55	971	1016	1159	1097	988	1111	1227	1085	905	1019	1074	971	840	1000	1059	986	895
60	679	822	814	864	690	884	860	857	657	819	748	815	592	823	741	811	626
65	474	700	609	736	490	763	632	720	489	696	578	690	434	694	573	692	457
70	377	576	510	614	389	639	532	587	397	574	504	549	371	550	484	557	379
75	342	407	420	436	349	445	431	416	337	407	404	372	335	382	404	391	342
80	273	287	350	313	283	297	332	280	253	274	313	256	238	277	328	273	258
85	150	171	236	199	167	187	218	165	134	152	171	136	118	146	179	146	132
90	29	30	48	47	43	47	49	36	23	18	12	8	9	13	15	17	19
95	2	1	2	2	2	1	3	2	1	2	2	2	2	2	2	1	1
100	1	2	1	2	2	1	3	1	1	2	2	2	1	2	2	2	2
105	1	2	2	2	1	2	1	1	2	2	2	1	2	2	2	2	2
110	1	2	2	1	1	2	2	1	2	1	1	2	2	2	1	2	2
115	2	2	2	1	2	2	2	2	2	2	2	2	2	2	2	2	1
120	2	2	2	2	3	2	2	2	2	2	3	2	2	2	2	2	2
125	2	3	3	3	2	2	3	2	2	3	3	2	2	3	3	2	3
130	2	3	3	3	2	2	2	2	3	3	3	3	3	3	3	3	3
135	3	3	3	3	3	4	3	3	4	3	3	4	4	3	3	3	3
140	4	4	3	4	4	4	4	4	4	4	4	4	4	4	4	4	4
145	5	5	5	4	4	5	5	4	4	5	4	5	4	5	4	5	5
150	4	4	4	4	4	4	5	5	5	5	5	5	5	5	5	5	4
155	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
160	6	5	5	6	6	5	5	6	6	6	6	6	6	5	6	5	6
165	7	7	7	6	6	7	6	6	6	7	7	7	6	6	6	7	7
170	8	8	8	8	7	7	7	8	8	8	8	8	7	7	7	7	8
175	9	9	8	8	7	8	8	9	9	9	9	8	7	8	8	8	9
180	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9

Corrected UGR values generated per CIE 190-2010

UGR Table - Corrected

Reflectances											
Ceiling Cavity		70	70	50	50	30	70	70	50	50	30
Walls		50	30	50	30	30	50	30	50	30	30
Floor Cavity		20	20	20	20	20	20	20	20	20	20
Room Size		UGR Viewed Crosswise					UGR Viewed Endwise				
X=2H	Y=2H	13.5	15.0	13.9	15.3	15.6	14.1	15.5	14.4	15.9	16.2
	3H	15.1	16.4	15.5	16.7	17.1	15.6	17.0	16.0	17.3	17.6
	4H	15.9	17.1	16.3	17.4	17.8	16.3	17.6	16.7	17.9	18.3
	6H	16.6	17.7	17.0	18.1	18.5	17.0	18.2	17.5	18.6	18.9
	8H	16.9	18.0	17.3	18.4	18.8	17.4	18.5	17.8	18.8	19.2
	12H	17.1	18.2	17.6	18.5	19.0	17.7	18.7	18.1	19.1	19.5
4H	2H	14.0	15.2	14.4	15.6	16.0	14.5	15.8	14.9	16.1	16.5
	3H	16.0	17.1	16.4	17.5	17.9	16.6	17.6	17.0	18.0	18.4
	4H	17.0	17.9	17.4	18.3	18.8	17.5	18.4	17.9	18.8	19.3
	6H	17.9	18.7	18.3	19.1	19.6	18.4	19.2	18.8	19.6	20.1
	8H	18.3	19.0	18.7	19.4	19.9	18.8	19.5	19.2	20.0	20.4
	12H	18.6	19.2	19.0	19.7	20.2	19.1	19.8	19.6	20.3	20.8
8H	4H	17.5	18.2	17.9	18.6	19.1	17.9	18.7	18.4	19.1	19.6
	6H	18.5	19.2	19.0	19.7	20.1	19.0	19.6	19.5	20.1	20.6
	8H	19.0	19.6	19.6	20.1	20.6	19.5	20.1	20.0	20.6	21.1
	12H	19.5	20.0	20.0	20.5	21.1	20.0	20.6	20.6	21.1	21.6
12H	4H	17.5	18.2	18.0	18.7	19.2	18.0	18.7	18.5	19.1	19.6
	6H	18.7	19.3	19.3	19.8	20.3	19.2	19.7	19.7	20.2	20.7
	8H	19.3	19.8	19.8	20.3	20.9	19.8	20.3	20.3	20.8	21.4

Maximum UGR = 21.6

THD and PF Test

Model No.	BPANEL-75S5-UNV-D-8SC61-24-G-WH-P	Sample ID.	7712178

Test Method

1. The samples were tested according to the ANSI C82.77-10-2014.
2. The ambient temperature condition was maintained at 25 °C ± 1 °C. The sample measurement was made using a digital power meter and power supply. The sample was operated at rated voltage and stabilized before measurement. The total harmonic distortion were calculated from the digital power meter.

Test Results

CCT Range	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Current THD	Orientation
0%-3000K	120	60	0.595	71.0	0.9949	8.91%	Horizontal
	277	60	0.259	70.4	0.9803	7.17%	Horizontal
20%-3500K	120	60	0.592	70.7	0.9948	9.06%	Horizontal
	277	60	0.258	70.0	0.9799	7.14%	Horizontal
40%-4000K	120	60	0.592	70.7	0.9948	9.13%	Horizontal
	277	60	0.258	70.0	0.9799	7.08%	Horizontal
60%-5000K	120	60	0.592	70.7	0.9948	9.08%	Horizontal
	277	60	0.258	70.0	0.9799	7.09%	Horizontal
80%-5700K	120	60	0.592	70.6	0.9948	9.02%	Horizontal
	277	60	0.257	69.9	0.9799	7.14%	Horizontal
100%-6500K	120	60	0.594	71.0	0.9948	9.01%	Horizontal
	277	60	0.259	70.2	0.9801	7.03%	Horizontal

In-Situ Temperature Measurement Test

Model No.	BPANEL-75S5-UNV-D-8SC61-24-G-WH-P	Sample ID.	7712178

Test Method

1. In-Situ Temperature Measurement Test is conducted according to the UL 1598-2008, Section 14.
2. The testing was conducted in a room with ambient temperature of 25 °C ± 5 °C. The apparatus construction followed those described in UL1598-2008 for normal temperature testing. Thermocouples were placed on the LED package in the locations indicated by LM-80 report. Thermocouples were placed on the LED driver case in the locations specified by the manufacture if necessary. The temperature was recorded after the lamp was operated by 7.5 hours.
3.The data and photos in LM-80 test report is provided by the customer. The data and photos in driver specification is provided by the customer.

In-Situ Temperature Measurement Test Conditions

Temperature (°C)	Voltage (Vac)	Frequency (Hz)	Current (A)	Power (W)	Power Factor	Orientation
23.9	120	60	0.409	48.9	0.9955	Horizontal

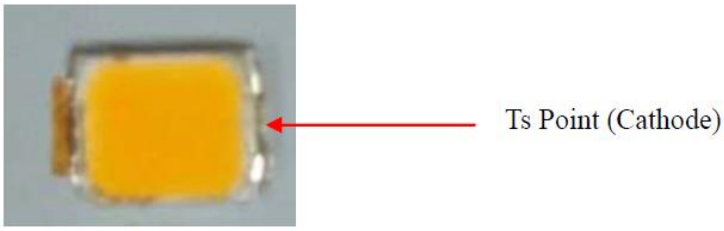
Test Results (LEDs)

Thermocouple Location	Manufacturer Declared Current (mA)	Temperature for Lighting source (°C)		LED Model Number	LM-80 Limit Current (mA)	LM-80 Limit Temp. (°C)
		Test result column 1	Test result (Correct to 25 °C)			
TMP of LEDs	100	37.5	38.6	BXVN-30E-13H-9NV	100	115
Ambient temperature	N/A	23.9	25.0			

Test Results(Driver)

Thermocouple Location	Temperature for Driver (°C)		Driver Model Number	Driver Limit Temp. (°C)
	Test result column 1	Test result (Correct to 25 °C)		
Tc of Driver	67.1	68.2	BQ-65L6-1470-44-M56	90
Ambient temperature	23.9	25.0		

TMP point in LM-80 report

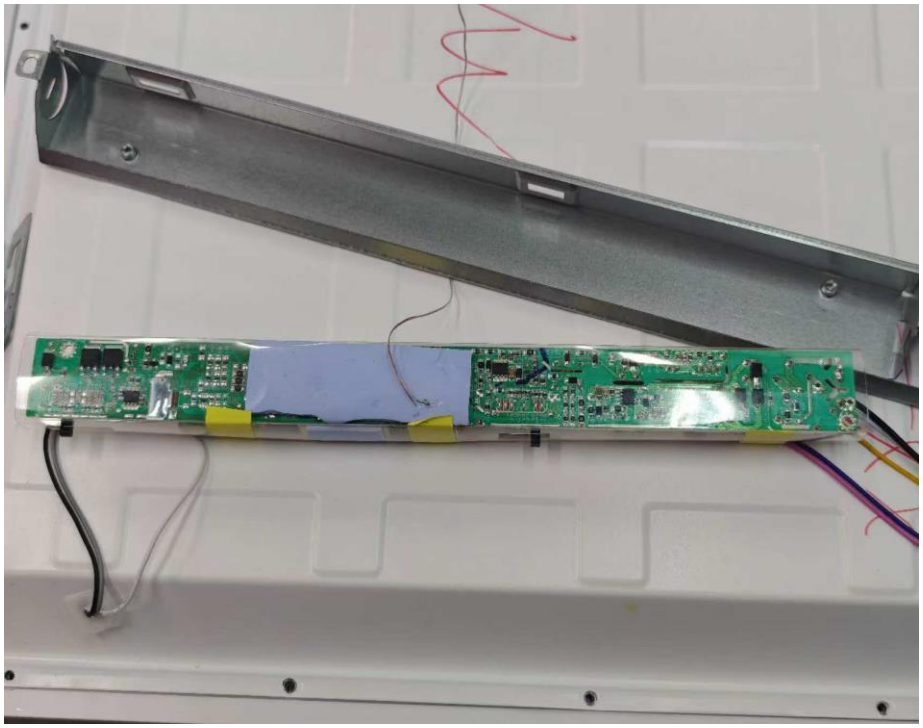


In-Situ Temperature Measurement Test (Cont'd)

Test Photos for TMP of LED Packages

In-Situ Temperature Measurement Test (Cont'd)

Test Photos for Tc Point of Driver



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